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Saito et al.

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(54) **GRAPEVINE PLANT NAMED 'SUNNY HEART'**

(50) Latin Name: ***Vitis labrusca* L. x *Vitis vinifera* L.**
Varietal Denomination: **SUNNY HEART**

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(30) **Foreign Application Priority Data**

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A01H 5/08 (2018.01)
A01H 6/88 (2018.01)

(52) **U.S. Cl.**
USPC **Plt./206**

(58) **Field of Classification Search**
USPC Plt./205, 206
See application file for complete search history.

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(57) **ABSTRACT**

A new variety of grapevine 'SUNNY HEART' that has green leaves and berries having a broad ellipsoid shape, red skin, and muscat flavor.

6 Drawing Sheets

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Common name: Grapevine.

Botanical classification: *Vitis labrusca* L. x *Vitis vinifera* L.

Variety denomination: 'SUNNY HEART'.

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority of Japanese plant variety rights application No. 37121 filed Nov. 13, 2023, the entire content of which is incorporated by reference herein.

BACKGROUND OF THE NEW VARIETY

The present invention relates to a new and distinct variety of grapevine, *Vitis labrusca* L. x *Vitis vinifera* L., which has been given the variety denomination 'SUNNY HEART'.

This new grapevine has green young leaves, R.H.S. Colour Chart: 148C, and berries having a broad ellipsoid shape, red skin, and muscat flavor, that distinguish it, notably from its male parent 'SHINE MUSCAT' (berries having yellow green skin) and female parent '626-84' (berries having dark red violet skin and sharp brisk flavor), as well as from reference varieties 'SEKIREI' (reddish young leaves, R.H.S. Colour Chart: 183B, 144A, berries having narrow ellipsoid shape and pale red skin) and 'KOTOPY' (greenish-red young leaves, R.H.S. Colour Chart: 148C, 178B, berries having narrow ellipsoid shape and pale red skin).

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ORIGIN OF THE VARIETY

The parent varieties are 'SHINE MUSCAT' (male parent) (not patented in the U.S.) and '626-84' (female parent) (not patented in the U.S.), respectively.

The variety was developed and propagated in Akitsu, Hiroshima, Japan.

The variety was reproduced asexually in Akitsu, Hiroshima, Japan by vegetative reproduction by grafting one-year-old cuttings on rootstocks.

In 2004, seedlings from crossing the parent varieties and collecting seeds were grown.

In 2007, they were grafted on root stock.

In 2009, the first fruit was harvested and lines were selected having fruit with good taste.

In 2016, plants were selected and named 'Akitsu 32' as temporary name for purposes of confidential testing for local adaptability.

In 2018-2022, growing tests were performed under confidentiality conditions at thirty-seven testing stations in Japan.

In 2022, budding and berry ripening times were measured.

On Nov. 13, 2023, an application for plant variety rights No. 37121 was filed for the present variety in the Japanese Plant Variety Protection Office of the Ministry of Agriculture, Forestry and Fisheries (MAFF), Japan.

Cultivation of the variety does not require special conditions or treatments.

The variety is currently mainly intended for fruit production (table grape).

SUMMARY OF THE VARIETY

The variety is distinguished by green young leaves, R.H.S. Colour Chart: 148C, and berries having a broad ellipsoid shape, red skin, and muscat flavor, that distinguish it, notably from its male parent 'SHINE MUSCAT' (berries having yellow green skin) and female parent '626-84' (berries having dark red violet skin and sharp brisk flavor), as well as from reference varieties 'SEKIREI' (reddish young leaves, R.H.S. Colour Chart: 183B, 144A, berries having narrow ellipsoid shape and pale red skin) and 'KOTOPY' (greenish-red young leaves, R.H.S. Colour Chart: 148C, 178B, berries having narrow ellipsoid shape and pale red skin).

A comparison with the male parent 'SHINE MUSCAT' is as follows (evaluation based on averages):

TABLE 1

No.	Characteristics	Male parent 'SHINE MUSCAT' States	Present variety 'SUNNY HEART' States
59	Mature leaf: depth of upper lateral sinuses	medium	shallow
63	Mature leaf: ratio length/width of teeth	medium	small
91	Berry: color of skin (without bloom)	yellow green (RHS 146C)	Red (RHS 187B)

A comparison with the female parent '626-84' is as follows (evaluation based on averages):

TABLE 2

No.	Characteristics	Female parent '626-84' States	Present variety 'SUNNY HEART' States
91	Berry: color of skin (without bloom)	dark red violet (RHS 184B)	Red (RHS 187B)
101	Berry: particular flavor	sharp brisk	Muscat

A comparison with the reference varieties 'SEKIREI' is as follows (evaluation based on averages):

TABLE 3

No.	Characteristics	Reference variety 'SEKIREI' States	Present variety 'SUNNY HEART' States
8	Young shoot: prostrate hairs on tip	absent or very sparse	very dense
11	Young leaf: color of upper side of blade	dark copper-red (RHS 183B, 144A)	Green (RHS 148C)
12	Young leaf: prostrate hairs between main veins on lower side of blade	absent or very sparse	very dense
57	Mature leaf: blistering of upper side of blade	weak	medium to strong

TABLE 3-continued

No.	Characteristics	Reference variety 'SEKIREI' States	Present variety 'SUNNY HEART' States
59	Mature leaf: depth of upper lateral sinuses	medium	shallow
90	Berry: shape	narrow ellipsoid	broad ellipsoid

A comparison with the reference variety 'KOTOPY' is as follows (average evaluation):

TABLE 4

No.	Characteristics	Reference variety 'KOTOPY' States	Present variety 'SUNNY HEART' States
11	Young leaf: color of upper side of blade	light copper-red (RHS 148C, 178B)	Green (RHS 148C)
12	Young leaf: prostrate hairs between main veins on lower side	dense	very dense
59	Mature leaf: depth of upper lateral sinuses	medium	shallow
66	Mature leaf: prostrate hairs between main veins on lower side	medium	dense
90	Berry: shape	narrow ellipsoid	broad ellipsoid

A comparison with both 'SEKIREI' and 'KOTOPY' is as follows (average evaluation):

TABLE 5

No.	Characteristics	Reference variety 'SEKIREI' States	Reference variety 'KOTOPY' States	Present variety 'SUNNY HEART' States
12	Young leaf: color of upper side of blade	dark copper-red (RHS 183B, 144A)	light copper-red (RHS 148C, 178B)	green (RHS 148C)
13	Young leaf: prostrate hairs between main veins on lower side of blade	absent or very sparse	dense	very dense
59	Mature leaf: depth of upper lateral sinuses	medium	medium	shallow
90	Berry: shape	narrow ellipsoid	narrow ellipsoid	broad ellipsoid

BRIEF DESCRIPTION OF THE DRAWINGS

In the accompanying drawings, which are as nearly true as is reasonable possible to make in a color illustration of this type:

FIG. **1** is a color photograph showing a general view of the grapevine named 'SUNNY HEART' bearing vine;

FIG. **2** is a color photograph showing a bunch of grapes on the vine of 'SUNNY HEART';

FIG. **3** is a color photograph showing flower clusters of 'SUNNY HEART';

FIG. **4** is a color photograph showing flowers of 'SUNNY HEART';

FIG. **5** is a color photograph showing the flowers and separated organs of 'SUNNY HEART';

FIG. **6** is a color photograph showing a bunch of grapes cut from a vine of 'SUNNY HEART';

FIG. **7** is a color photograph showing berries of 'SUNNY HEART';

FIG. **8** is a color photograph showing cross-sections of berries of 'SUNNY HEART';

FIG. **9** is a color photograph showing berry shapes of 'SUNNY HEART' as well as 'SEKIREI' and 'KOTOPY' for comparison purposes;

FIG. **10** is a color photograph showing the upper and lower sides of young leaves of 'SUNNY HEART' as well as 'SEKIREI' and 'KOTOPY' for comparison purposes;

FIG. **11** is a color photograph showing the lower sides of mature leaves of 'SUNNY HEART' as well as 'SEKIREI' and 'KOTOPY' for comparison purposes.

Due to chemical and/or digital development, processing and printing, the plants or portions of plants depicted in the photographs may or may not be precisely accurate, when compared to the actual botanical specimens.

DETAILED DESCRIPTION OF THE INVENTION

The botanical description and the photographs are of the 'SUNNY HEART' plants evaluated in 2022 (including for the comparison with the reference plants 'SEKIREI' and 'KOTOPY'. Values as provided are averages.

The plants shown on the photographs were grown at Akitsu, Hiroshima, Japan. The plant is a 5-year grown plant. The young leaf and flower were photographed on May 29, 2023, the mature leaf was photographed on Jun. 29, 2023, and the bunch and berry was photographed on Sep. 15, 2023.

Colors are given according to The R.H.S. (Royal Horticultural Society) Color Chart, Sixth Edition (2015).

BOTANICAL DESCRIPTION

TABLE 6

No.	Characteristics	States	Values
	Tree		
1	Average height		180 cm
2	Average width		60 cm
3	Vigor	strong	
4	Trunk diameter (at a given height above soil line)		19.8 mm (at 20 cm above soil line)
5	Grown on rootstock	Yes	Teleki selectia Kober 5BB
6	Time of bud burst	early	early April (approx. Apr. 7)
	Young shoot		
7	Openness of tip	wide open	
8	Prostrate hairs on tip	very dense	
9	Anthocyanin coloration of prostrate hairs on tip	absent or very weak	
10	Erect hairs on tip	absent or very sparse	
	Young leaf		
11	Color of upper side of blade	green	148C (RHS)
12	Prostrate hairs between main veins on lower side of blade	very dense	

TABLE 6-continued

No.	Characteristics	States	Values
13	Erect hairs on main veins on lower side of blade	absent or very sparse	
	Shoot		
14	Average length of stem		186 cm
15	Average diameter of stem		22.0 mm
16	Texture of stem	rough	
17	Color of stem	brown	N200C, 200C (RHS)
18	Average length of branches		88 cm
19	Average diameter of branches		16.0 mm
20	Texture of branches	intermediate	
21	Color of branches	grey-brown	199C, N199C (RHS)
22	Average length of shoots		5.9 m
23	Average diameter of shoots		9.8 mm
24	Texture of shoots	smooth	
25	Color of shoots	green	144A (RHS)
26	Attitude (before tying)	semi-erect	
27	Color of dorsal side of internodes	green and red	144A, 183B (RHS)
28	Color of ventral side of internodes	green	144A (RHS)
29	Shoot: color of dorsal side of nodes	green and red	144A, 183B (RHS)
30	Shoot: color of ventral side of nodes	green	144A (RHS)
31	Erect hairs on internodes	absent or very sparse	
32	Length of tendrils	medium	19.4 cm
	Flower		
33	Average size of flower buds		Width 1.5 mm, Length 2 mm
34	Color of flower buds	green	144A (RHS)
35	Inflorescence	panicle	
36	Average number of petals on the flower		5
37	Color of petals (top surface)	green	144A (RHS)
38	Color of petals (bottom surface)	green	144B (RHS)
39	Average petal size		Width 0.4 mm, Length 2 mm
40	Length of flower clusters	medium	19.3 cm
	Sexual organs		
41	Sexual organs	fully developed stamens and fully developed gynoecium	
42	Average size of seeds		Width 6.0 mm, Length 8.3 mm
43	Average number of seeds		2.4
44	Average color of seeds	greyed-orange	165B (RHS)
	Mature leaf		
45	Leaf arrangement	Alternate	
46	Color of leaf (top surface)	green	137B (RHS)

TABLE 6-continued

No.	Characteristics	States	Values
47	Color of leaf (bottom surface)	yellow-green	148C (RHS)
48	Shape of leaf	wedged-shaped	
49	Texture of leaf (top surface)	medium to strong blistering	
50	Texture of leaf (bottom surface)	rough	
51	Vein color	yellow-	145C (RHS)
52	Average length of petiole	green	121 mm
53	Average diameter of petiole		3.9 mm
54	Color of petiole	yellow-green and greyed-purple	145B(RHS), 183D(RHS)
55	Size of blade	Small to medium	267 cm^2
56	Shape of blade	wedged-shaped	
57	Blistering of upper side of blade	medium to strong	
58	Number of lobes	five	
59	Depth of upper lateral sinuses	shallow	3.4 cm
60	Arrangement of lobes of upper lateral sinuses	open	
61	Arrangement of lobes of petiole sinus	slightly open	
62	Length of teeth	medium	4.8 mm
63	Ratio length/width of teeth	small	0.56
64	Shape of teeth	mixture of both sides straight and both sides convex	
65	Proportion of main veins on upper side of blade with anthocyanin coloration	absent or very low	
66	Prostrate hairs between main veins on lower side of blade	dense	
67	Erect hairs on main veins on lower side of blade	absent or very sparse	
68	Length of petiole compared to length of middle vein	much shorter	0.55
	Fruit ripening		
69	Date of first pick in specific location	approx. Sep. 5	
70	Date of last pick in specific location	approx. Sep. 20	
71	Time of beginning of berry ripening	medium to late	early August (approx. Aug. 5)
72	Pounds per bushels per acre		About 890 pounds/acre
73	Storability of fruit	12 days	
	Bunch		
74	Average bunches per plant		5
75	Average length of bunch	large	183 mm
76	Average width of bunch		72 mm
77	Average weight of bunch		222.2 g
78	Average number of grapes per bunch		27.2
79	Average length of peduncle	medium	58 mm
80	Average diameter of peduncle		3.2 mm
81	Average color of peduncle	light red	144A, 184D (RHS)
82	Size (peduncle excluded)	large	183 mm
83	Density	medium	
84	Length of peduncle primary bunch	medium	58 mm
85	Color of peduncle	light red	144A, 184D (RHS)
	Berry		
86	Average size of berry		Width 24.8 mm, Length 26.0 mm
87	Color of berry skin	red	187B (RHS)
88	Color of berry flesh		144D (RHS)
89	Size	large	9.5 g
90	Shape	broad ellipsoid	
91	Color of skin (without bloom)	red	187B (RHS)
92	Bloom	many	
93	Ease of detachment from pedicel	medium	
94	Thickness of skin	thin to medium	
95	separation of skin and flesh	difficult	
96	Anthocyanin coloration of flesh	absent or very weak	
97	Firmness of flesh	medium	
98	Character of flesh	collapse	
99	Sweetness of flesh juice	high	20.7%
100	Juiciness of flesh	very juicy	
101	Particular flavor	Muscat	
102	Formation of seeds	complete	
	Woody shoot		
103	Main color	orange brown	177B (RHS)
	Ecotype		
104	Flower shatter	medium	

Other features of the plant are as follows:

Disease resistance.—Normal resistance was observed for pests and diseases in Akitsu, Hiroshima, Japan. Fungicide treatment was considered necessary against the following diseases: *Peronospora* (*Plasmopara*), *Oidium, Botrytis.*

I/we claim:

1. A new and distinct variety of grapevine named 'SUNNY HEART', substantially as described and illustrated herein.

* * * * *

FIG. 1

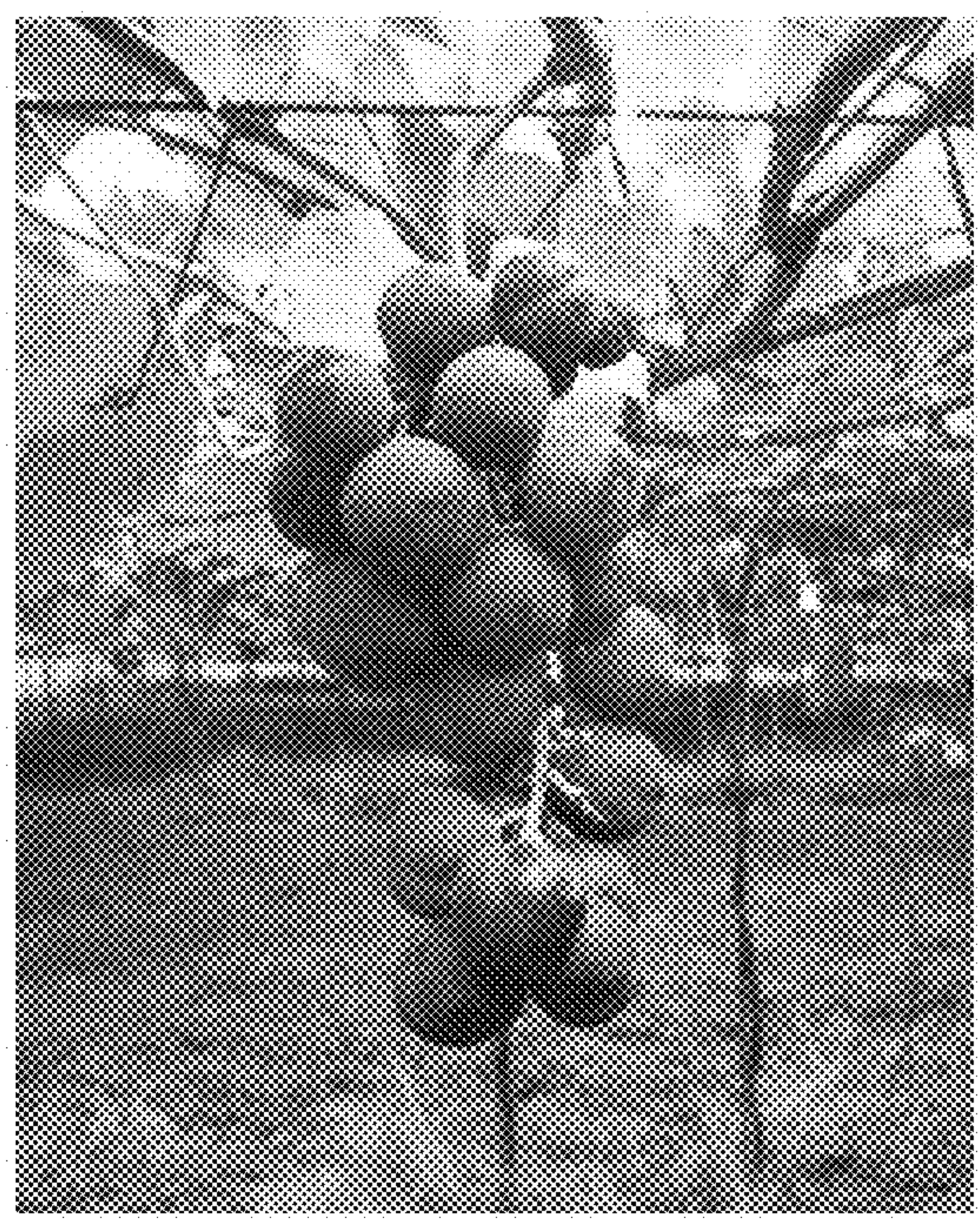

FIG. 2

FIG. 3

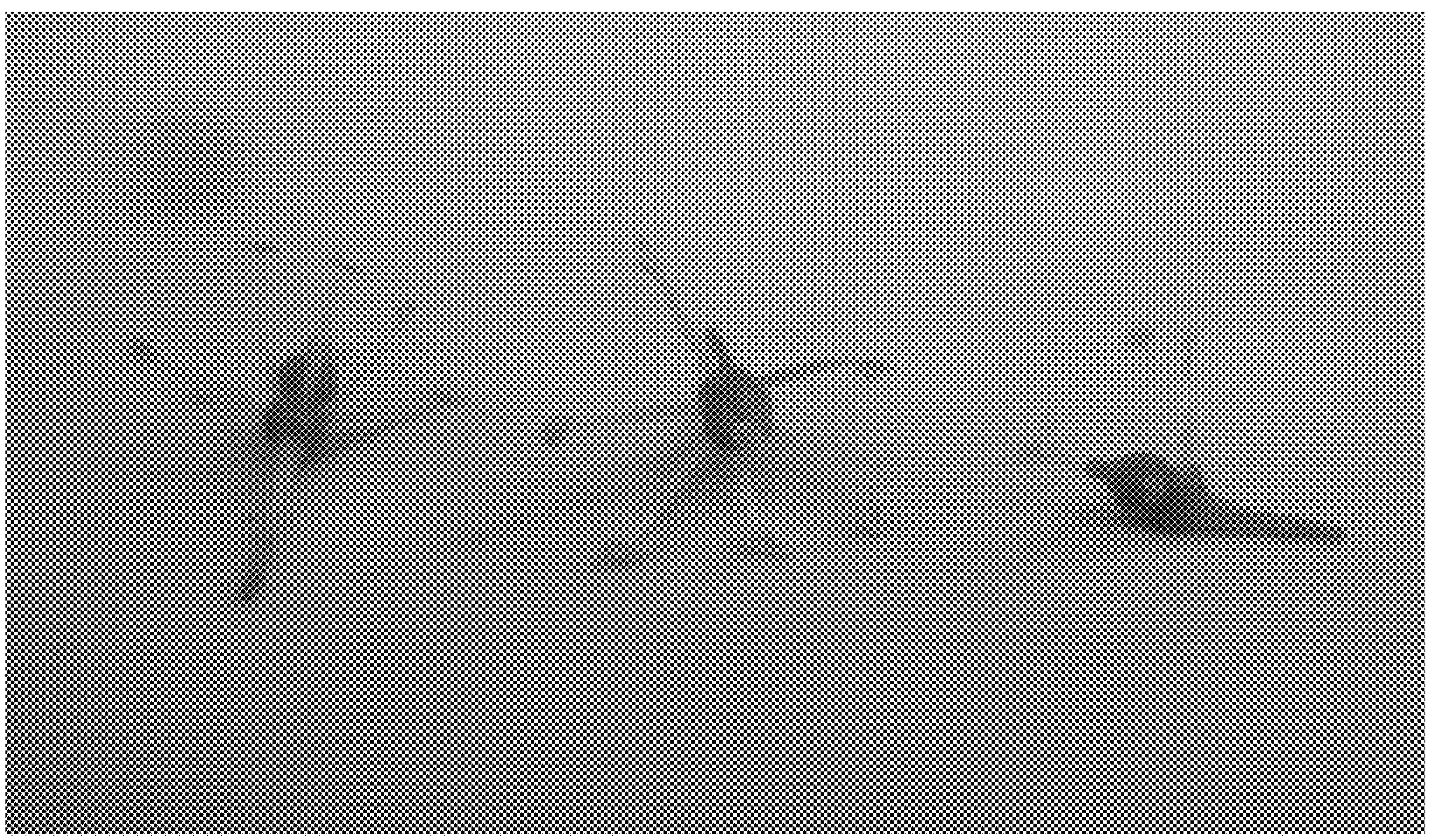

FIG. 4

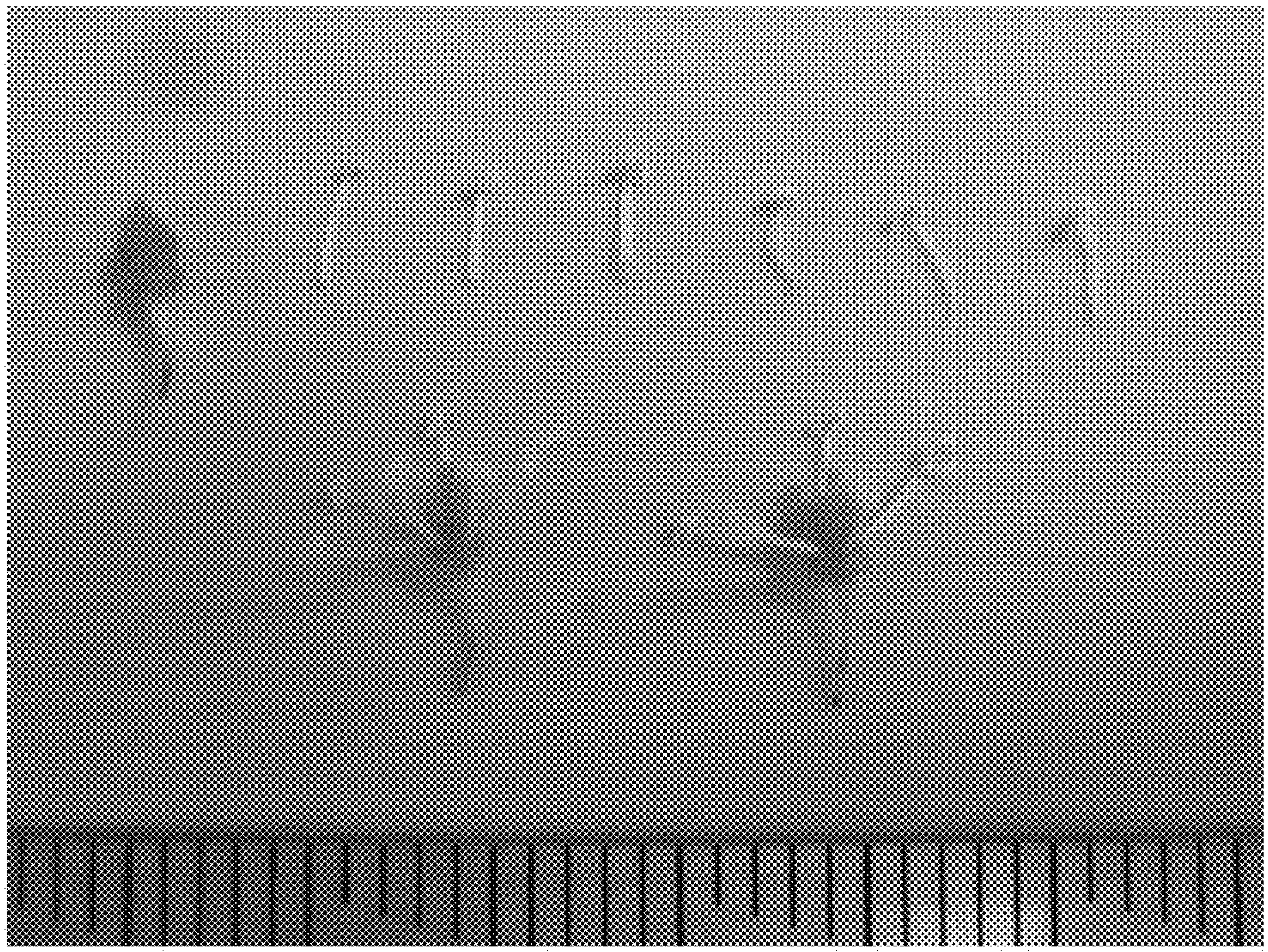

FIG. 5

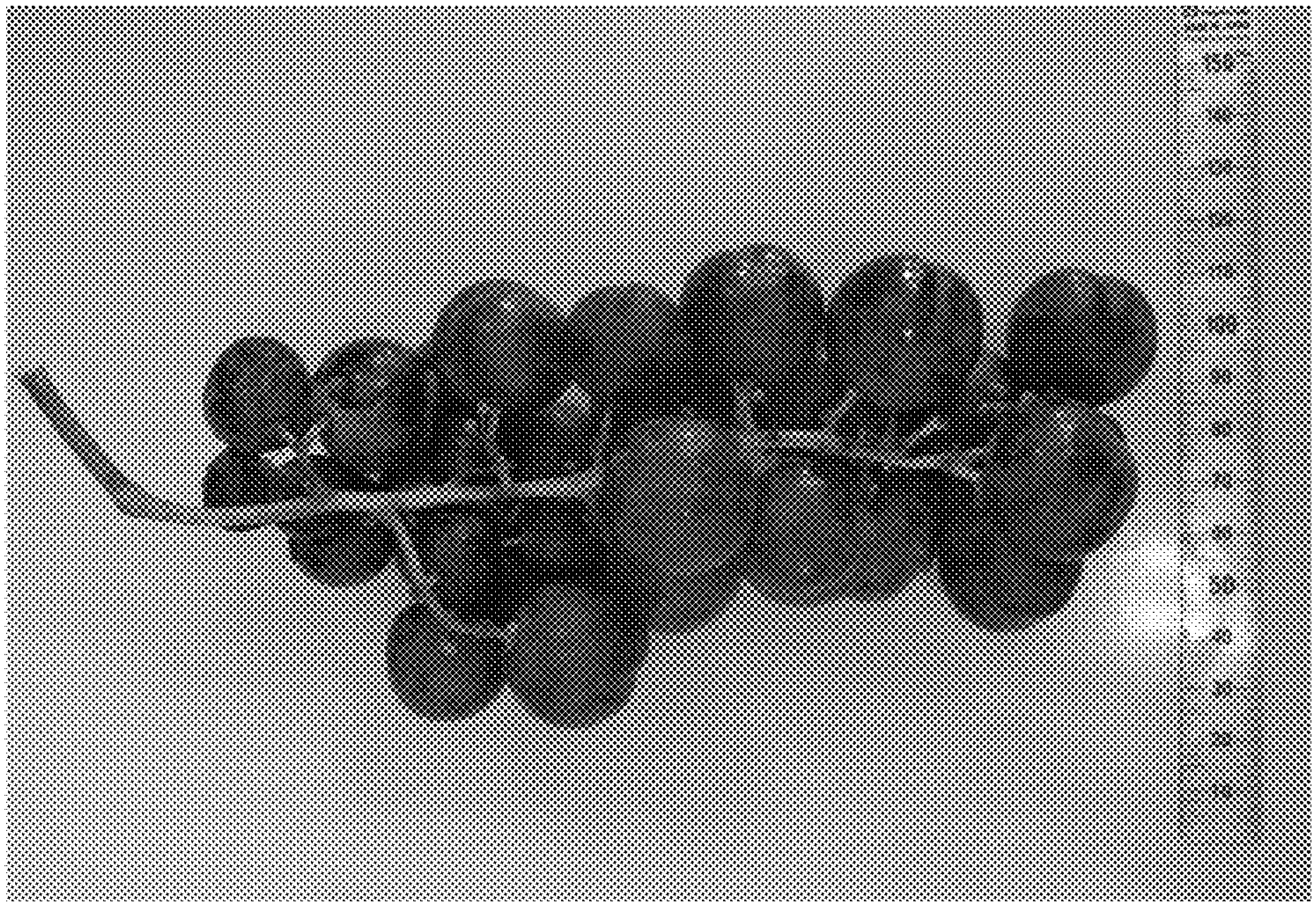

FIG. 6

FIG. 7

FIG. 8

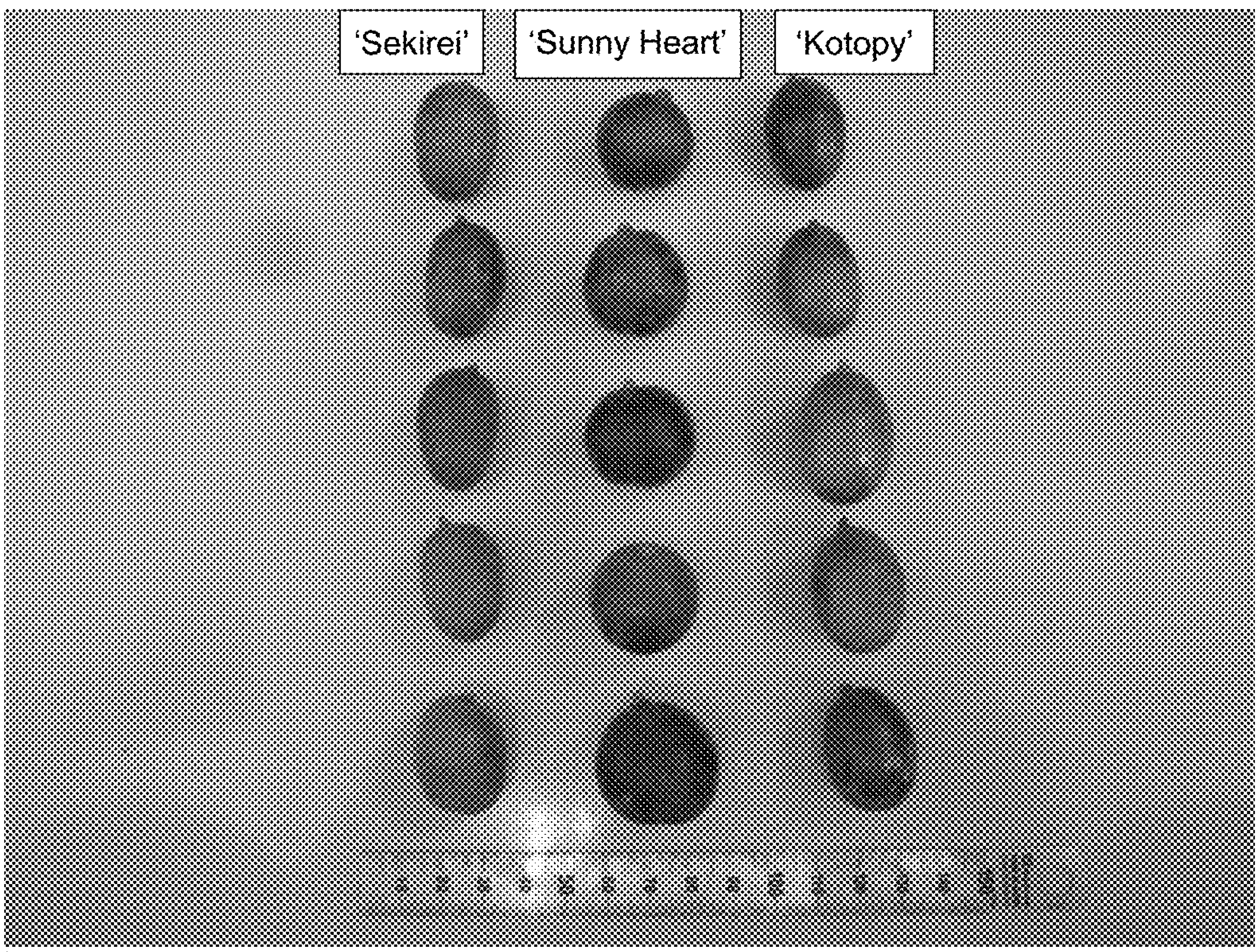

FIG. 9

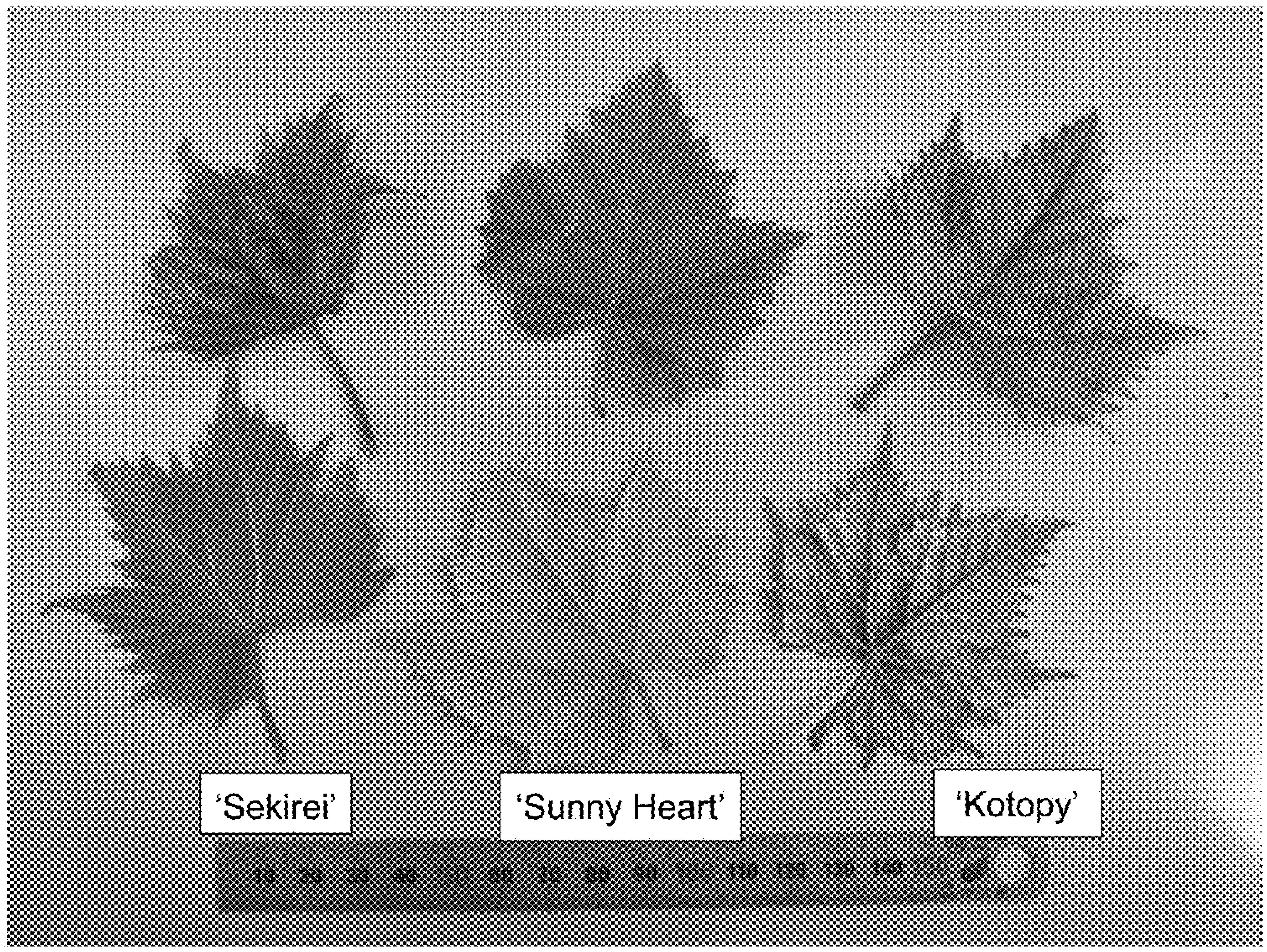

FIG. 10

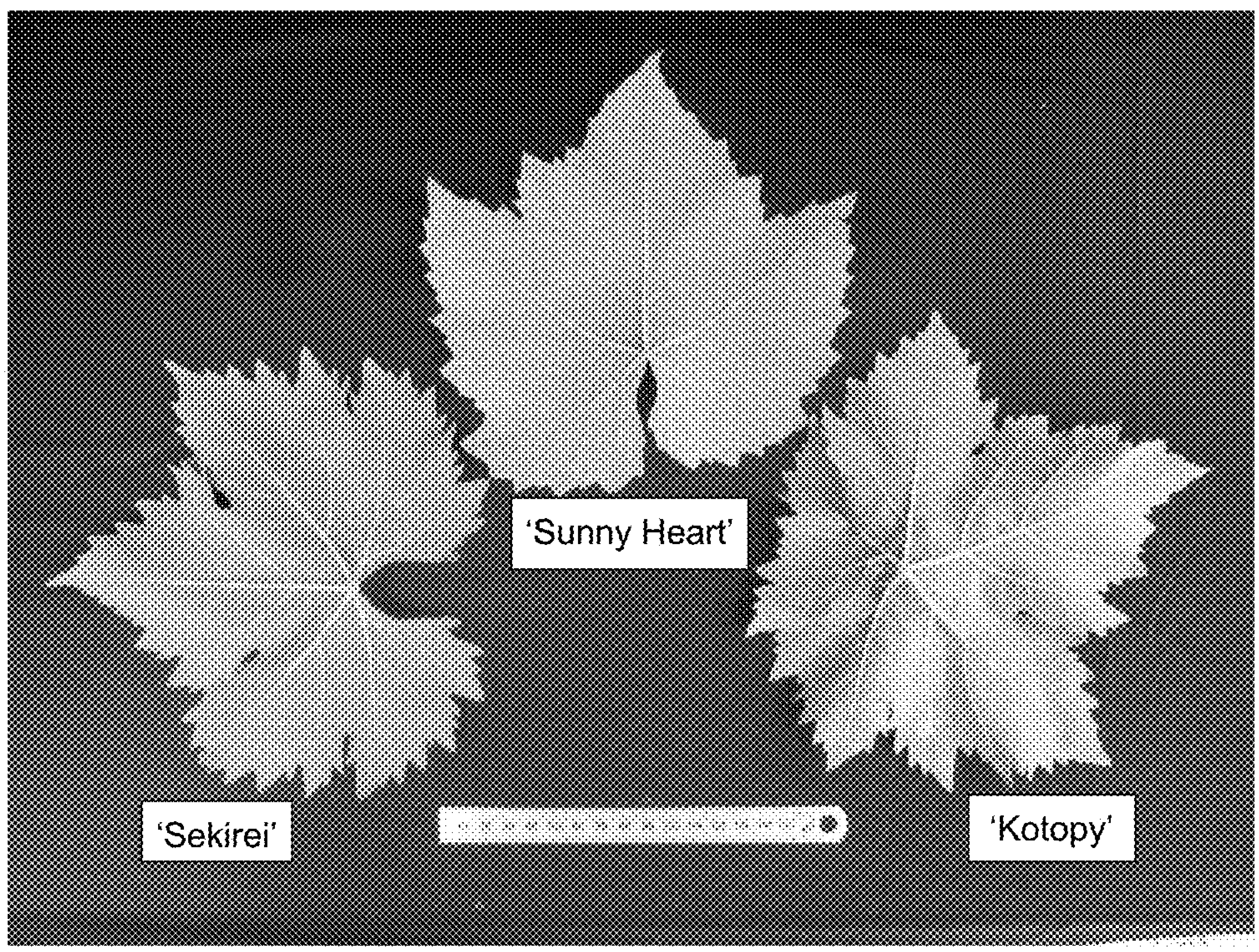

FIG. 11